P04

Ultrafast and Memory-Efficient Alignment of Short DNA Reads

Authors B. Langmead, C. Trapnell, M. Pop, S. L. Salzberg

Published Genome Biology (2009)

Paper Link https://doi.org/10.1186/gb-2009-10-3-r25

Domain DOM_BIO

Abstract

Bowtie aligns short sequencing reads to the human genome at rates exceeding 25 million reads per hour. It creates an FM-index from the Burrows-Wheeler Transform, indexing 3 billion base pairs in under 1.3 GB of RAM.

Abstract quoted from B. Langmead et al., Genome Biology (2009).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?